
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

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ABSTRACT

Algorithms increasingly mediate information across settings that range from social media feeds and targeted advertising to LLM-based search, advice, and assistance. In such environments, individual decisions often combine private, algorithmically mediated signals with observational learning from the actions of others. We study this broader class of interactions through a model of controlled sequential social learning, in which an information-mediating planner (e.g., an LLM or related algorithmic system) chooses the information structure faced by each agent while agents also learn from earlier decisions. The planner may aim to improve social welfare (an altruistic planner) or to induce a particular preferred action (a biased planner). This formulation yields an optimization problem for social learning that links dynamic programming, decentralized action choices, and Bayesian belief updates. Within this setting, we prove convexity of the value function and characterize optimal policies for altruistic and biased planners, identifying the tradeoffs between information-control costs and the payoffs generated by induced choices. The characterization shows that optimal mediation can move among qualitatively different regimes as public belief changes: maximal investment, no investment, increasing investment, or decreasing investment. In some belief regions, a biased planner optimally reduces signal precision, thereby strategically obfuscating agents’ information. Even under stringent transparency constraints—information parity with individuals, no lying or cherry-picking, and full observability—information mediation can substantially shift social welfare in either direction. We complement the theory with simulations in which LLMs instantiate both planner and agents. In these experiments, the LLM planner displays emergent strategic behavior that broadly mirrors the analytical trends, while deviations point to non-Bayesian reasoning patterns consistent with both human users and LLMs trained on human-like data. Overall, the framework provides a tractable basis for analyzing LLM and algorithmic information mediators in socially coupled decision environments.

1 INTRODUCTION

Large Language Models (LLMs) increasingly operate as information mediators across socially consequential domains, including search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) as well as personal assistants that provide medical, political, or other high-stakes advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). These deployments are promising, but LLMs have also shown persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) capacities in user interactions. At scale, such influence rarely acts on isolated individuals alone. It interacts with ordinary social transmission, as people observe one another and update their own decisions accordingly, creating population-level dynamics that can either amplify or dampen algorithmic effects.

As a running example, consider an LLM-powered recommendation system that advises a sequence of potential customers about a new restaurant. The restaurant’s quality—the payoff-relevant state—is either good or bad, and neither the users nor the system initially knows which state holds. Each user receives a personalized recommendation, interpreted as a private signal, and then chooses whether to patronize the restaurant. The recommendation system, acting as the *planner*, can ex-

pend resources to make that signal more informative; for instance, it may generate a more carefully tailored review using the user’s preferences. The same abstraction also covers other settings in which a mediator can improve the precision of user-facing information through market research, surveys, data acquisition, or personalization.

A user’s decision, however, is not determined by the private signal alone. Users can also observe earlier choices—for example, if many predecessors patronized the restaurant, the next user may infer that they likely received favorable information. This form of *social learning* shapes behavior in many domains, including vaccine adoption (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), technology adoption (e.g. Gillingham & Bollinger (2021); Weber (2012)), and political or moral opinion formation (e.g., Brady et al. (2021); Guilbeault et al. (2018)). Our central contribution is to analyze the strategic interaction between a planner that controls private information and a population of agents that learns socially. The planner must therefore account not only for its immediate effect on the current user, but also for the information *externality* that the user’s action creates for all future users.

Although real-world platforms and user populations are more complex than any stylized model, we isolate a broadly applicable mechanism through a tractable model of *controlled social learning*. In the model, an algorithmic planner strategically chooses, at a cost, the precision of private signals shown to sequential agents. The planner’s influence is deliberately constrained: it does not falsify information, but decides how much to invest in making signals clearer and more informative. This abstraction lets us ask general questions about mediated social learning: how can a planner steer collective beliefs, how should it adapt to evolving public opinion, and what are the resulting welfare consequences?

The answers depend on the planner’s objective, a distinction that applies beyond the restaurant example. An *altruistic* planner, such as an ideal recommendation system, seeks to maximize user utility by helping each agent choose the action that matches the state (patronize the restaurant if and only if it is good). A *biased* planner instead seeks to induce a particular action, such as patronizing the restaurant, regardless of the true state.

This objective distinction is related to, but not identical with, alignment. By definition, an altruistic planner is aligned with maximizing the agent’s *expected utility* under uncertainty. A biased planner’s objective is state-independent and therefore not aligned in that ex ante sense, yet its actions need not be harmful in every realized state. If the state favors the planner’s preferred action (e.g., the restaurant is in fact good), the planner’s influence can coincide with the agent’s *realized utility*. We therefore study both altruistic and biased mediation across cases where planner and agent interests are aligned or misaligned.

Contributions and Outline This paper makes the following contributions to the broader study of mediated social learning:

1. **A Theoretical Framework for Controlled Social Learning.** We introduce a formal model that combines a dynamic control problem for a centralized information mediator with sequential social learning among agents. The planner strategically chooses the precision of private signals to pursue an objective that may be altruistic (maximizing social welfare) or biased (inducing a specific action).
2. **A Characterization of Optimal Planner Policies.** We characterize optimal policies for altruistic and biased planners as functions of evolving public belief. For the altruistic case, the result relies on a proof of value-function convexity. The characterizations identify general tradeoffs that arise whenever present information choices create informational externalities for later agents.
3. **LLM-based Evaluation of Strategic Mediation.** We conduct simulations in which LLMs act as both planner and agents. The experiments show that (a) accounting for social learning can greatly change public opinion and social welfare relative to myopic control, (b) emergent LLM-planner strategies broadly align with the theoretical policy structure even against non-Bayesian LLM agents, and (c) LLMs display strategic reasoning that is relevant for both beneficial information support and societal risk.

We review related work in Section 2, introduce the formal model in Section 3, and derive optimal policies in Sections 4 and 5. We then evaluate the theory with LLM-based simulations in Section 6 and conclude in Section 7. Proofs, extensions, and experimental details appear in the appendices.

2 RELATED WORK

Our work connects controlled information mediation to several broader literatures: social learning, information design, online persuasion and RL, and LLM-based strategic interaction.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) introduced the canonical sequential social-learning framework in economics, and later work has characterized these dynamics in more general environments (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). This literature also examines richer forms of social information and a variety of biases and misperceptions among agents (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010) as well as Bikhchandani et al. (2024) and the references therein).

Work on the *control* of social learning is more limited (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The closest papers are Wei & Anastasopoulos (2022) and Smith et al. (2021). Wei & Anastasopoulos (2022) allows two-way communication between planner and agents, so agents can report private signals in exchange for action recommendations. Smith et al. (2021) studies a benevolent planner that directly changes agents’ choice rules and also proposes a decentralized pivot mechanism to improve learning. Our model instead preserves unilateral agent choice and does not require private reports to the planner, matching a broader class of information-mediating algorithms that influence decisions through the information they present rather than through direct control of actions.

Information Design In our setting, the planner optimizes its objective by choosing the precision of agents’ private signals, giving a constrained instance of classical information design (Bergemann & Morris, 2019). This literature rose to prominence with Kamenica & Gentzkow (2011)’s Bayesian persuasion model, in which a sender chooses the signal structure observed by a rational Bayesian receiver facing an unknown state. Here, each period is a costly and constrained information-design problem: the planner chooses a binary symmetric signal at some cost. The main additional feature is that receivers are socially linked, so each period’s information choice can affect later beliefs through observed actions.

Few papers study information design directly within sequential social-learning environments (Arieli et al., 2022; Wu et al., 2025). Both choose an information structure once at the outset and then hold it fixed for the sequence of agents. Arieli et al. (2022) studies a self-interested sender with a general information structure, as in classical Bayesian persuasion. Wu et al. (2025) restrict the information structure to be more informative in the Blackwell sense and maximize the probability of a correct cascade. We instead study a dynamic setting in which the sender can choose a new structure for each agent, allowing strategies that respond to the evolving public belief.

Online Persuasion and RL Recent work also connects online learning with Bayesian persuasion and information design. These papers study online sequential persuasion with unknown environmental parameters, where a sender repeatedly interacts with short-lived receivers without knowing the prior over states and/or receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A related line studies sequential persuasion in dynamic environments or MDPs, introducing state transitions and/or receiver foresight; these models emphasize history-dependent or promise-based signaling policies and often connect to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). The central goals in these works are typically learning-based guarantees, such as sublinear regret relative to the best signaling scheme in hindsight, or approximation algorithms under computational constraints.

By contrast, our planner need not learn unknown environmental parameters. Sequential dependence comes from social learning among receivers, not from the sender learning the state of the environment. The planner is also not informationally advantaged: it observes neither the underlying state nor private signals. These restrictions allow us to characterize optimal policies for the planner. Although the problem is sequential, it is not an online-learning or RL environment in the usual sense: actions do not alter the payoff-relevant state, and the planner does not explore to learn unknown

parameters. Instead, the planner observes agents' actions as signals of social belief propagation, so intertemporal dependence arises from information externalities generated by those actions.

LLMs A growing set of papers studies LLMs in information-design and persuasion settings (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These works focus on one-on-one interactions and therefore abstract away from the social dynamics central to our setting. The closest is Duetting et al. (2025), who study information design where a sender chooses both a signal structure and a framing that conditions a possibly non-Bayesian receiver's posterior belief. They use an LLM-based oracle to handle the computationally difficult optimization over framing, building on evidence that LLMs can mimic humans as economic agents (Dillion et al., 2023; Horton, 2023). Their model does not study repeated interaction or social learning, but we adapt parts of their methodology for the empirical evaluation in Section 6.

3 SOCIAL LEARNING MODEL

We consider a planner interacting with a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature draws a fixed, *unknown* exogenous state $\omega \in \Omega := \{G, B\}$, with $\mathbb{P}(\omega = G) = b_1$, and the prior b_1 is common knowledge. In the recommendation example from Section 1, G denotes the state in which the restaurant is good, but the same binary-state abstraction represents any setting with two payoff-relevant alternatives.

At each time i , the planner chooses how much to invest in the information shown to agent i . It then provides a private signal $s_i \in \Omega$ that supports the agent's decision. We model this signal as binary and symmetric: it matches the state ω with probability $q_i \in [0.5, 1]$, i.e. $\mathbb{P}(s_i = \omega) = q_i$. We call q_i agent i 's signal precision.

Conditioned on ω , each agent's signal is independent of other agents' signals and of the history. Agent i combines this signal with the observed actions of previous agents $j < i$ and chooses an action $a_i \in \Omega$. The agent receives utility 0 when her action matches the state ($a_i = \omega$) and utility $-C$ otherwise, where $C > 0$. Each agent maximizes her own expected utility. In the running example, this means patronizing the business exactly when the state is G , but the same payoff structure captures any binary decision with a costly mismatch.

3.1 AGENTS' DECISION PROBLEMS

Each agent observes all predecessor actions and the signal precisions attached to those actions. We denote this public history by $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because the planner and all agents share the same history $\mathcal{H}_i$, they also share a *public* belief about ω that updates after each action. Immediately before agent i acts, this belief is $b_i = \mathbb{P}(\omega = G | \mathcal{H}_i)$; before any action is observed, it is the prior b_1 . As in the classic models of Banerjee (1992) and Bikhchandani et al. (1992), $(b_i)_{i \in \mathbb{N}}$ is a Markov process and a sufficient statistic for $\mathcal{H}_i$ (see Appendix B.3).

Given $\mathcal{H}_i$ (equivalently, by the Markov property, b_i) and q_i , agent i chooses $a_i \in \Omega$ to maximize expected utility. If both actions yield the same expected utility, she breaks ties by choosing the action matching s_i . Combining the public history with her private signal of precision q_i , agent i forms the following *private* belief $\tilde{b}_i$ about ω (derived in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

The agent then chooses the action corresponding to the more likely state under $\tilde{b}_i$: $a_i = G$ if $\tilde{b}_i > 0.5$, $a_i = B$ if $\tilde{b}_i < 0.5$, and $a_i = s_i$ when $\tilde{b}_i = 0.5$. Substituting for $\tilde{b}_i$ gives the decision rule

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Subsequent agents observe agent i 's action and incorporate it into the next public belief b_{i+1} as follows (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, agent i 's action perfectly reveals her private signal through Equation (2), so the updated public belief equals her private belief. Outside this region, Equation (2) implies that the private signal does not affect the action; the action therefore conveys no new information and the public belief remains fixed. The process has then reached an absorbing state, called an information cascade or herding in classic social-learning models such as Bikhchandani et al. (1992); Banerjee (1992). At such points, the public belief no longer learns from agents' private information.

Under this decision rule, agent i 's expected utility is $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$, which reduces to the following expression (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We study two planner objectives that capture aligned and state-independent forms of mediation. An altruistic planner seeks to induce correct actions ($a_i = \omega$), whereas a biased planner seeks to induce a particular action, say G , regardless of ω . We use subscripts A and B for these planners. In both cases, the planner controls each agent's private-signal precision. The cost of a chosen precision is represented by $\beta(\cdot)$, a non-negative, increasing, continuous, and concave function of its argument. The planner's information set is the same public history $\mathcal{H}_i$ available to the agents.

Altruistic Planner At time i , the altruistic planner chooses the precision q_i for agent i and pays $\beta(q_i)$, where $\beta(p) = 0$ for a baseline $p \in [0.5, 1)$. Because lowering precision is never beneficial for the altruistic objective (see Appendix C.9), we normalize costs so that the planner pays only for increases above the baseline. Additional precision is more costly, with decreasing marginal costs. Agents know both p and $\beta(\cdot)$.

The altruistic planner maximizes social welfare, defined as agents' expected total utility, net of precision-investment costs. Let $r_A(b_i, q_i)$ denote the planner's instantaneous reward when the public belief is b_i and the chosen signal precision for agent i is q_i . Starting from b_1 and following policies $\pi = (\pi_i)_{i=1}^\infty$ with $\pi_i(\mathcal{H}_i) = q_i$, the planner obtains the following expected total discounted utility for discount factor $\delta \in [0, 1)$:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The altruistic planner's optimal utility and policy are the supremum and arg supremum of $V_A^\pi(\cdot)$ over $\pi \in \Pi$, where Π denotes the policy class. Setting $\delta = 0$ recovers the corresponding myopic value and policy.

Biased Planner The biased planner differs from the altruistic planner in objective, and hence in the associated reward and cost structure. Appendix A gives a political-campaign example, but the same formulation applies whenever a mediator prefers one action independently of the state. The biased planner seeks to induce action G from each agent regardless of ω . If an agent chooses G (respectively, B), the planner incurs cost 0 (respectively, $C > 0$), independent of the true state. Unlike the altruistic planner, the biased planner may either increase or decrease precision relative to the baseline p , and either deviation is costly. Any precision other than p requires costly tailoring of the information shown to the agent, such as learning how that agent best interprets a message. The biased planner therefore pays $\beta(|q_i - p|)$ when choosing precision q_i , with $\beta(0) = 0$.

The biased planner's expected instantaneous reward at time i is

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

Using Equation (6), we define $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ for the biased planner analogously to $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ under Equation (5).

As Section 3 specifies, each agent still prefers the action that matches ω . Therefore, when $\omega = B$, a successful biased planner lowers the agent's utility by inducing G . By contrast, the altruistic planner always seeks actions that match ω , so its success raises agent utility. Thus, the altruistic objective is aligned with each agent's objective in all states, whereas the biased objective is aligned only when $\omega = G$.

For both objectives, the planner's problem is an infinite-horizon discounted stationary Markov Decision Process (MDP) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition function given by Equation (3) (Puterman, 1990). A unique optimal value function therefore exists (Kallenberg, 2011). We focus on deterministic Markov policies, so $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$.

Remark 2. *The model makes three assumptions that delimit, but do not trivialize, the scope of the results: (1) The planner observes the same history as the agents. This excludes mediators with richer private data, but covers settings in which all parties rely on the same public evidence, such as early findings or survey data. (2) The planner can change agents' signal precision, while signals still pass through a binary symmetric channel. This excludes cherry-picking, censorship, and falsification, but covers mediators whose role is to improve the quality of noisy information without changing its content. (3) The planner's control choices are fully observable. This excludes covert targeting or framing, but matches more transparent environments in which precision or information quality is visible.*

4 OPTIMAL ALTRUISTIC POLICIES

We first analyze the myopic benchmark, which abstracts away from the intertemporal effects created by social learning. In our formulation, this corresponds to setting the discount factor to $\delta = 0$, so the planner optimizes only the current agent's outcome and cost. When $\delta = 0$, $V_A^\pi(b) = r_A(b, \pi(b))$, and the optimal myopic policy $\pi_A^0(\cdot)$ is

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

This myopic altruistic benchmark can also be viewed as a decentralized precision-choice problem in which each agent selects and pays for her own private-signal precision to optimize only her current expected payoff net of cost.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimum therefore has a simple threshold structure. When public belief is already strong, the planner chooses the zero-cost baseline precision p ; when public belief is weak, it provides a perfect signal of precision 1. The threshold depends only on the cost of a perfect signal, $\beta(1)$, relative to the cost C of an incorrect action. If $\beta(1) \geq C$, perfect information is too expensive to justify for a single agent, so $t_M = 0.5$ and the belief region in which the myopic policy chooses precision 1 is empty.

We next give a structural result for the altruistic value function that supports the broader dynamic characterization:

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2 (Appendix C.3) is involved and may be useful beyond this particular application. The main difficulty is that agents' actions depend endogenously on the public belief. If actions were independent of the belief process, as in settings such as Nyarko (1994), expected utility would be linear in the belief state and value-function convexity would follow directly.

This convexity result is the key tool for characterizing the dynamic optimal policy.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & o.w. \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal policy separates the public-belief space into three regimes. First, as in the myopic benchmark, the dynamic optimum does not invest in additional precision when public belief is extreme. The no-investment region is narrower than in the myopic case, however, because the dynamic planner internalizes how today's action affects future agents through social learning.

When public belief is close to 0.5 and therefore weakly informative, the dynamic optimum selects precision 1 whenever the cost is not prohibitive. By Equation (2), the agent's action then equals the true state with probability 1, causing public belief to collapse to either 0 or 1.

In the intermediate regime, the dynamic optimum chooses the minimum precision $\max(b, 1 - b)$ that still makes the agent's action reveal her private signal (see Equation (2)). Any lower precision would make the action uninformative beyond existing public belief. Thus, the policy selects the lowest-cost precision that keeps observational social learning active.

5 OPTIMAL BIASED POLICIES

We again begin with the myopic benchmark, now for a biased planner. As in Section 4, the myopic biased policy $\pi_B^0(\cdot)$ maximizes only the instantaneous reward $r_B(b, q)$. The characterization below shows that the biased case has a richer structure: an optimal policy need not always exist (Puterman, 1990, sec. 2.3.1). Where the optimum is not attained, we use ϵ -optimal policies, denoted with an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

- (A) *If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.*
- (B) *If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.*
- (C) *If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.*
- (D) *If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

The proof appears in Appendix C.6. We then use Theorem 4 to characterize the dynamic optimal biased policy.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon, *}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5 (proven in Appendix C.8 and illustrated in figure 2a) identifies five belief-dependent regimes.

When public belief is sufficiently strong, as in (A), the cost of steering the system can exceed the planner's benefit, even when $b < 0.5$ is unfavorable to the planner. The chosen precision p is then below $\max(b, 1 - b)$, so Equation (2) implies that the agent follows public belief regardless of her private signal. The action is uninformative, public belief does not move, and the Markovian policy repeats this outcome indefinitely. This is an unfavorable cascade when $b < 0.5$ and a favorable cascade otherwise.

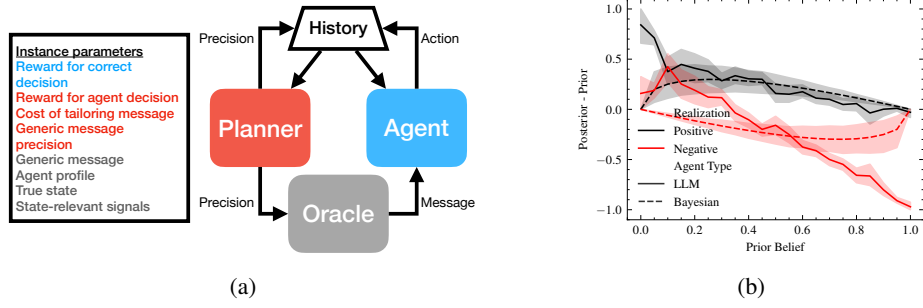


Figure 1: (a) Experimental system diagram. Instance parameters are color-coded by the LLM role that uses them. (b) Change in LLM-agent beliefs (solid) and Bayesian beliefs (dashed) from a fixed prior after a positive (black) or negative (red) signal.

When public belief is close to an unfavorable cascade, as in (B), the planner raises precision enough for the private signal to affect the action, even though the expected signal is B . The agent then follows her private signal, preserving a nonzero chance of the planner-favored action. This regime represents a last opportunity to steer the system away from an unfavorable cascade.

For slightly higher beliefs, as in (C), the planner may reduce precision below p . Because $b < 0.5$ in this region, more precise signals are more likely to produce news unfavorable to the planner. The planner therefore keeps precision high enough to influence the action ($q \geq 1 - b$), hoping for movement toward a favorable belief, but uses the least precise signal that can still affect behavior.

When public belief weakly favors the desired action, as in (D), the planner chooses precision at least p . Since $b > 0.5$, higher precision makes the agent more likely to infer $\omega = G$, so investment in this regime reinforces a favorable public belief.

When public belief is higher still, as in (E), the planner lowers precision just below $\max(b, 1 - b)$ to $b - \epsilon$. Agents then ignore private signals and choose G . In this region, the risk that a private signal disrupts favorable public belief outweighs both the cost of reducing precision and the possible benefit of further strengthening belief.

6 EVALUATION VIA LLM-BASED SIMULATION

We next use LLM-based simulations to examine how the theoretical mechanisms behave when both mediators and agents are instantiated by modern language models. The evaluation has three parts: we identify deviations from Bayesian updating among LLM agents, examine how an LLM planner adapts its strategy in response, and measure the resulting effects on social welfare.

To operationalize controlled social learning, we instantiate the model in a car-purchase scenario in which agents decide whether to buy a new car. This scenario preserves the binary-state, binary-action structure while providing natural language content for the LLMs. As shown in figure 1a, LLMs serve in three roles, with prompts reported in Appendix E.2.2:

Agent: Role-playing as an assigned profile, each agent observes previous actions and a private message about the car with known precision, forms a belief about car quality, and decides whether to buy.

Planner: After observing the action history, the planner selects the precision q_i of agent i 's private signal according to its objective.

Oracle: Given the planner's precision choice, an agent profile, and a car fact sheet (see Appendix E.2.1), the oracle generates a tailored private signal with the desired precision. We compare these LLM simulations with numerical evaluations of the planner's MDP under the same model parameters. In both LLM and numerical experiments, the cost function is linear, $\beta(q) = k|q - p|$; we vary k , baseline precision p , and discount factor δ , while fixing the cost of an incorrect action at $C = 1$. Appendix E gives the experimental setup and prompts, and Appendix E.3 validates both reported beliefs and oracle performance.

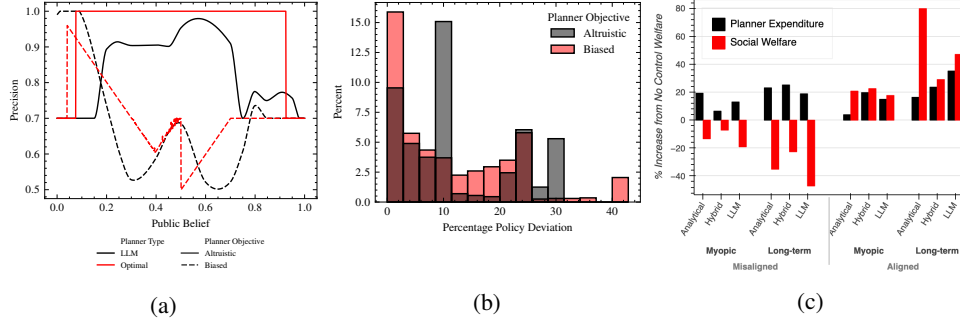


Figure 2: (a) Representative policies from the LLM planner (black) and analytically optimal planner (red) under altruistic (solid) and biased (dashed) objectives. (b) Distribution of percentage deviations between LLM and optimal policies. (c) Planner expenditure and social-welfare change, measured as percentages of no-control baseline welfare. The true state is fixed to B; the left half shows a biased planner seeking action G, and the right half shows an altruistic planner.

Our experiments support three findings about information mediation in socially coupled environments:

1. **The importance and impact of social learning control:** Analytical and LLM planners can substantially improve their own utility by accounting for social-learning dynamics rather than treating agents independently. Depending on whether planner and agent objectives are aligned, the same mechanism can improve or harm social welfare.
2. **The robustness of our analytical characterization:** LLM planners choose policies that are structurally similar to the non-obvious analytical optima despite interacting with non-Bayesian LLM agents.
3. **The emergent strategic behavior of LLM:** LLM planners exhibit strategic behavior in their policy choices. Their policies are close to the analytical policies in broad structure, and their deviations appear to account for the non-Bayesian behavior of the LLM agents they face.

6.1 LLMs AS NON-BAYESIAN AGENTS

The analytical results assume Bayesian agents, while both humans and LLMs are known to exhibit systematic cognitive biases (Rupprecht et al., 2025). To understand how this behavioral mismatch affects the mediated-learning setting, we first test how an LLM updates beliefs after receiving new information. The results in figure 1b identify non-Bayesian LLM updating patterns that are relevant for downstream social learning (see also Appendix E.4):

- (NB1) LLM agents underreact to private signals that align with their prior beliefs.
- (NB2) LLM agents overreact to private signals that run counter to their prior beliefs.
- (NB3) As a result, LLM agents require stronger public belief before entering an information cascade.

Related cognitive patterns have also been observed empirically in human studies (e.g., Ba et al. (2022); Chan et al. (2025)). The subsequent results show that the LLM planner’s policy remains structurally close to the theoretical policy even when agents display these non-Bayesian patterns.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We next compare LLM-planner policies with the optimal policies derived in Sections 4 and 5. As shown in figure 2a, the emergent LLM strategies retain substantial structural similarity to the theoretical optima despite non-Bayesian agent behavior. Thus, even when the optimal policy is non-trivial (e.g., the biased policy in figure 2a), the LLM planner uses the social-learning structure rather than merely optimizing each interaction in isolation.

For altruistic objectives, both planners stop investing when public belief is strong and invest heavily when public belief is uncertain. For biased objectives, both planners show the same qualitative regimes: high investment to escape an unfavorable belief, reduced precision near the midpoint, and no investment once favorable belief is established. The deviations are often modest, as shown in figure 2b: for both planner types, deviation is less than 10% for the majority of belief states. This supports the broader structural alignment between the LLM strategy and the analytical characterization.

The remaining structural differences are consistent with strategic adaptation to the non-Bayesian behaviors identified in Section 6.1. (1) The LLM planner tends to avoid extreme precisions (0.5 or 1.0), consistent with a known central tendency bias (Rupprecht et al., 2025). (2) The LLM planner tapers investment more gradually rather than using a sharp cutoff. This is consistent with agents’ resistance to cascades (NB3): even agents with strong priors may be swayed by a countervailing private signal, so stopping investment is less safe than in the Bayesian model. (3) The biased LLM planner also continues investing at very low beliefs, consistent with the possibility that agents overreact (NB2) to a surprisingly positive signal, making a last-ditch intervention more viable than in the Bayesian case.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because better information never harms an agent’s expected utility, social welfare is monotone in signal precision (see Appendix C.9). The altruistic planner therefore improves social welfare relative to the baseline. To quantify the magnitude of these effects across behavioral assumptions, we compare social welfare and planner expenditure in three settings: (1) the *analytical* setting (optimal policy, Bayesian agents), (2) the *LLM* setting (LLM planner, LLM agents), and (3) a *hybrid* setting (optimal policy, LLM agents).

The results in figure 2c show that planners in all evaluated settings can materially alter social welfare. Neglecting social learning, as in the myopic cases, also substantially worsens planner outcomes. In particular, the biased analytical and LLM planners decrease social welfare by 40 to 50% when misaligned. They do so by intentionally obscuring information about the true state through policies that lower signal precision. The size of this welfare loss is notable because the planners operate under the restrictive transparency assumptions in Remark 2, highlighting the risk posed by misaligned information mediators even when their control channel is limited.

The results also illustrate the cost of misspecifying agent behavior. In the hybrid setting, the analytically optimal policy is designed for Bayes-rational agents and becomes brittle when applied to non-Bayesian LLM agents. The LLM policy, by contrast, remains close to the analytical policy in broad structure while adapting more effectively to agents with human-like biases.

7 CONCLUSION

We introduced a framework for studying strategic control of social learning by an algorithmic information mediator. Within the model’s transparent and binary-signal setting, even a constrained planner can substantially affect social welfare in either direction, and accounting for social learning is central to using this influence effectively. The LLM simulations reinforce these mechanisms: LLM planners exhibit strategic reasoning that broadly mirrors the analytical policy structure while adapting to non-Bayesian LLM agents.

Taken together, the results suggest several broader lessons for algorithmic information mediation. First, human-AI and human-algorithm interactions should not be studied only as isolated dyads when users also learn from one another; social learning can change both beneficial and harmful impacts. Second, information mediators can guide or derail public belief even through the relatively subtle channel of precision choice, motivating further study of safeguards. Third, modern LLMs can exhibit strategic behavior that accounts for both social learning and non-Bayesian cognitive biases. These points underscore the need to understand how risks from algorithmic mediators, and LLM mediators in particular, can be mitigated. A limitation of our study is the absence of human-subject data. Because the fidelity of LLM-human simulators remains debated (see, e.g., Horton (2023); Gao et al. (2025)), human data would be valuable for validating or calibrating the LLM simulations.

Two future directions are especially important: generalizing the model and mitigating the negative welfare effects of biased planners. Several extensions would broaden the formal scope while relaxing current assumptions. Larger state spaces, more general signal structures, richer loss functions, and heterogeneous agents would bring the model closer to many applications. Moving beyond binary states and 0-1 losses appears primarily algebraic, and we conjecture that the qualitative policy regimes will persist. For more general signal structures, the main challenge is preserving the convexity result of Theorem 2; identifying the broadest signal classes for which such convexity holds is an important open problem. As an initial step toward heterogeneity, Appendix D extends the altruistic-planner results to agents with opposing preferences.

Another important direction is preventing welfare losses from biased planners. Regulatory or mechanism-design approaches that better align planner and agent incentives may help limit harmful welfare effects while preserving beneficial information mediation.

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A APPLICATIONS ELABORATED

As an altruistic-mediation example, consider a sequence of individuals deciding whether to patronize a business or service provider, such as a restaurant, contractor, or realtor, while a recommendation system provides information about that provider. The binary unknown state is whether the provider is good. Both the recommendation system and the agents prefer actions that match this state: agents want to patronize the provider if and only if it is good, and the system wants to induce the same state-contingent behavior. In this sense, the recommendation system is an *altruistic planner*.

The system can show an agent a highly targeted, and hence *precise*, message that presents the provider’s strengths and weaknesses in a context suited to the agent’s background. It can also show a generic or confusing message that is less useful for decision-making. Higher precision makes the signal more informative, comprehensible, and relatable, increasing the chance that the agent reaches the correct conclusion. Social welfare rises when more agents make correct decisions, but changing message precision is costly because it requires tailoring to agent-specific characteristics. The recommendation system therefore chooses precisions to maximize social welfare net of tailoring costs.

For a biased-mediation example, consider voters who are motivated to support the candidate most likely to win in their community. The incentive to “back a winner” has been shown to affect electoral outcomes, helping explain why US states with early primary elections can have outsized influence (see Bartels (1988)). The planner is a particular candidate’s campaign, which wants agents to back that candidate regardless of the true state. The binary state indicates whether the candidate is winning. If an agent knew the state, she would back the candidate if and only if he were winning.

Each agent understands her community well enough to infer whether the candidate is winning if she accurately understands his positions and policies. That understanding depends on how information is presented. Some

agents may respond best to audiovisual material, others to long-form articles with facts and citations, others to brief focused summaries, and others to content in a native language.

The campaign sends digital ads, interpreted as private signals, with varying precision. Precision captures how well the content is tailored to the agent’s way of processing information. A precise signal accurately conveys the state by helping the agent understand the candidate’s policies, record, and character, thereby enabling her to infer whether he will win. Precision does not necessarily help the campaign: if the candidate is losing in the agent’s community, a highly precise signal would induce opposition. An imprecise signal can instead make the state harder to infer, for example by being vague or confusing, and may increase the chance that the agent believes the candidate is winning. A biased planner can therefore prefer to decrease precision when the true state is unfavorable. Tailoring either precise or imprecise ads requires costly knowledge of how the agent understands content; for example, confusing an agent who prefers brief summaries may require deliberately sending verbose long-form material. The planner chooses precisions to maximize the expected number of backers minus the costs of generating those precisions.

B SOCIAL LEARNING DYNAMICS

We first summarize the sequence of developments at step i :

1. The planner determines q_i based on $\mathcal{H}_i$ and, q_i is observed by all agents.
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i based on the observable history and their private signal with the aim of maximizing their own utility.
4. All other agents (and the planner) observe a_i and the public belief at the end of step i , b_{i+1} is updated accordingly.

Note that from (1), q_i is a function of $\mathcal{H}_i$. Thus, if a random variable is conditioned upon both q_i and $\mathcal{H}_i$, the conditioning on q_i is redundant. We will use this principle throughout.

B.1 PRIVATE BELIEF UPDATE

We now describe how agent i chooses action a_i . Let $\alpha_i := \frac{b_i}{1-b_i}$ be the public likelihood ratio based on observed actions. Agent i forms a private belief about ω , $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$, by combining the current public belief with her private signal, s_i . We consider a private likelihood ratio, $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. We now describe how $\tilde{\alpha}_i$ and $\tilde{b}_i$ are computed.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \tag{8}$$

$$= \frac{q_i}{1-q_i} \alpha_i \tag{9}$$

In obtaining Equation (8), we apply the assumption that, conditioned upon ω , agents’ signals are independent of one another and the history to assert that $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$. The same is done when conditioning upon $\omega = B$. We additionally use the fact that q_i is a function $\mathcal{H}_i$.

2. If private signal is $s_i = B$, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \tag{10}$$

We let $y(b, q)$ and $z(b, q)$ denote the probability of realizing signal G or B , with public belief b and signal precision q conditioned upon the history. By conditioning on ω and applying the Law of Total Probability,

these probabilities can be expressed as follows:

$$\begin{aligned}
y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\
&= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\
&= q_i b_i + (1 - q_i)(1 - b_i) = 1 + 2q_i b_i - q_i - b_i
\end{aligned} \tag{11}$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \tag{12}$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields the following:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1 - q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \tag{13}$$

Note that if $s_i = G$ then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ cannot be 0. Similarly, if $s_i = B$, then $z(b_i, q_i)$ cannot be 0. Thus, the equation above is well-defined.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i 's action a_i will be G if $\tilde{b}_i$ exceeds 0.5 (equivalently, if $\tilde{\alpha}_i > 1$), B if $\tilde{b}_i < 0.5$ (equivalently, if $\tilde{\alpha}_i < 1$), and s_i otherwise as per the tie-breaking rule. This action and i 's precision q_i are observed by all peers, resulting in an updated public belief b_i .

When $b_i < 1 - q_i$, then $\alpha_i < \frac{1 - q_i}{q_i}$. Therefore, if $s_i = G$, $\tilde{\alpha}_i < 1$ by Equation (9); if $s_i = B$, $\tilde{\alpha}_i < \frac{(1 - q_i)^2}{q_i^2} \leq 1$ (since $q_i \geq 0.5$) by Equation (10). Thus, no matter the private signal realization s_i , $a_i = B$ when $b_i < 1 - q_i$. Similar arguments can be used to show that when $b_i > q_i$, $a_i = G$. Intuitively these cases are when the strength of the public belief overpowers the precision of the private signal.

Now consider the case that $1 - q_i \leq b_i \leq q_i$. Then if $s_i = G$, $\tilde{\alpha}_i \geq 1$ from Equation (8) and since $b_i \geq 1 - q_i$. If $s_i = B$, then $\tilde{\alpha}_i \leq 1$ from Equation (10) and since $b_i \leq q_i$. Together with the tie-breaking rule, this results in action $a_i = s_i$. This leads to Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Thus, agent i 's action is only informative in the first case of Equation (2) where it allows future agents to perfectly infer i 's private signal. Thus, in this case, public belief will update just as the private belief from Equation (13). In the latter two cases of Equation (2), the action contains no information about the true state. Thus, the public belief will not change. This yields Equation (3) reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \tag{14}$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now prove that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, regardless of the rest of $\mathcal{H}_i$. It follows then that

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \tag{15}$$

From Equation (2), for the second and third cases, it is clear that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, regardless of the rest of $\mathcal{H}_i$. We consider the first case next.

$$\begin{aligned}
&\mathbb{P}(a_i = G | \mathcal{H}_i, q_i) \\
&= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\
&= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\
&\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\
&= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\
&\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\
&= b_i q_i + (1 - b_i)(1 - q_i)
\end{aligned} \tag{16}$$

The first step applies the law of total probability. In the second step, using the fact that q_i is a function of $\mathcal{H}_i$, we substitute b_i for $\mathbb{P}(\omega = w|\mathcal{H}_i, q_i)$. In the third step, we use the fact that, for $1 - q_i \leq b_i \leq q_i$, $a_i = s_i$, and, further, that s_i is independent of the history when conditioned upon ω . The result follows.

This leads to the following theorem:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G|b_i, q_i, a_i\} \quad (17)$$

In other words, b_i captures the entire information pertinent to the update of b_{i+1} that is contained in $\mathcal{H}_i$.

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G|b_i, q_i\}}{\mathbb{P}\{a_i|b_i, q_i\}}$$

We will prove the claim via induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G|b_1, q_1\} \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\} \mathbb{P}\{\omega = G|b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G|\mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\} \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G|\mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\} \mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G|b_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

Note that we have shown that the public belief is Markovian adapting the arguments in Banerjee (1992) and Bikhchandani et al. (1992). The modifications were required because of the differences in selections of the agents' precisions caused by the introduction of the central planner.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We prove this for each case of a_i delineated in Equation (2). In the second case, we can reason as follows:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq G|b_i, q_i) = 1 - b_i$$

Similarly, in the third case of Equation (2), we obtain:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq B|b_i, q_i) = b_i$$

For the case when $1 - q_i \leq b_i \leq q_i$ and $a_i = s_i$ from Equation (2), we can reason as follows:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1-q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) allows us to reason as follows

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As stated in Section 3.2, we can restrict the altruistic planner to precision $q \in [p, 1]$ because social welfare is increasing in precision (see Appendix C.9). Accordingly, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) leads to optimal precision p if $p < \max(b, 1 - b)$. That is, if the signal is not precise enough to influence the agent's action, then the planner will not expend resources for it. In the latter case, because $\beta(\cdot)$ is increasing and concave, $\dot{\beta}(\cdot)$ is positive and decreasing. Thus, $r_A(b, q)$, if the optimal lies in this range it must be attained at $q = 1$.

We now identify the regime in which the latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, the above inequality is never satisfied, and it is never optimal to select precision 1. Otherwise, the inequality above yields $b \in \left[\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C} \right]$. This yields Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We inductively prove that the expected k -th stage reward, i.e., the expected utility of the planner from the control and action of the k -th agent, is convex. The instantaneous reward (Equation (5)), which is convex with respect to public belief, provides our base case.

The first challenge encountered is the unusual nature of the public belief update. Although the state space is uncountably infinite, the belief update only takes support on a maximum of 2 values. To manage this, we define a decision tree, i.e., the complete binary tree of all possible trajectories once an initial belief and policy are fixed. Each node has two children corresponding to each possible signal realization the next agent might receive. The root is the expected instantaneous reward at time 1, i.e., from the first agent's action. The induction moves down the levels of this tree with the k -th level containing 2^{k-1} nodes, each associated with a sequence of realizations of $k - 1$ signals.

We then show that for a node in the $(k - 1)$ -th level that has convex expected reward, its two children in the k -th level satisfy the same property. This is where we must deal with the dependence of agents' actions on public belief. Note that even when applying the same precision and receiving the same signal realization, two agents beginning at different public beliefs may take opposing actions (see Equation (2)). Thus, standard results that provide easy ways of bounding the future terms of the Markov process do not apply. Here, our specific belief update is actually helpful. We can leverage the fact that Bayesian updates are martingales (i.e., $\mathbb{E}[b_{i+1}] = b_i$). Along with the convexity of instantaneous rewards, this allows us to complete the inductive step and, subsequently, the proof.

Preliminaries We first prove the following lemma containing a few useful properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1 - b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_A^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned}
V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right]
\end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now introduce the concept of *decision trees* for Markovian and more general policies. A policy can be represented by a decision tree, which is a complete binary tree as in figure 3. Every node of the decision tree represents the state of the system, i.e., the corresponding belief value when the controller sends its private signal to the agent corresponding to the level of the node. The branches represent the actions taken by the agent corresponding to the level once it receives its private signal. The controller's policies (i.e. choice of precision) determine the probabilities of the actions from each node and the belief resulting from the action. The probabilities of an action at a certain node can be considered the weight of the branch. A decision tree is uniquely identified by the belief values associated with nodes and the weights of the branches. For a deterministic stationary Markovian policy, the controller's choice of precision, and therefore the probability associated with an action, is a deterministic function of the belief value of the node, regardless of the level of the node in the tree and the path to the node from the root. Thus, the decision tree of any such policy, i.e. the belief values at the nodes and the weights of the branches, is determined entirely by the value of the initial state. For an arbitrary policy of the controller, the controller's decision at a given node can depend on the entire path leading to the node.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We will first show that, for fixed precision, the instantaneous reward r_A is convex in public belief. We can rewrite r_A as follows using Lemma 7:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave. As we are subtracting the minimum, the overall function is convex with respect to x .

We consider arbitrary x_0, m_0, n_0 such that $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. To show the convexity of $V_A^*(\cdot)$ we must show that $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. By definition of the optimal value function, $V_A^*(x) \geq V_A^\pi(x)$ for any policy π . Thus, it will suffice to show that there exist $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

We refer to the decision tree of π^* when the initial state is x_0 as Λ .

We consider $\tilde{\pi}$ to be a policy of the controller that chooses the precision at each node to be the same as what Λ does at the counterpart node. The nodes of decision trees are considered to be counterparts if they can be reached via the same sequence of actions from the root (see figure 3 for illustration). Thus, the precision chosen by this policy is the same as what Λ does after the same set of actions of the agents. We refer to the decision trees for $\tilde{\pi}$ with initial state m_0 (respectively, n_0) as Λ_m (Λ_n).

We now illustrate the policy $\tilde{\pi}$ for initial values m_0 and n_0 considering the decision trees Λ_m and Λ_n . While doing so, we illustrate counterpart nodes and the weights of the branches. We will select one path of actions λ out of these decision trees by specifying set $\mathcal{G}$ as the epochs where action G is taken and $\mathcal{B}$ as its complement. Refer to figure 3 for illustration. Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the

public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n . Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

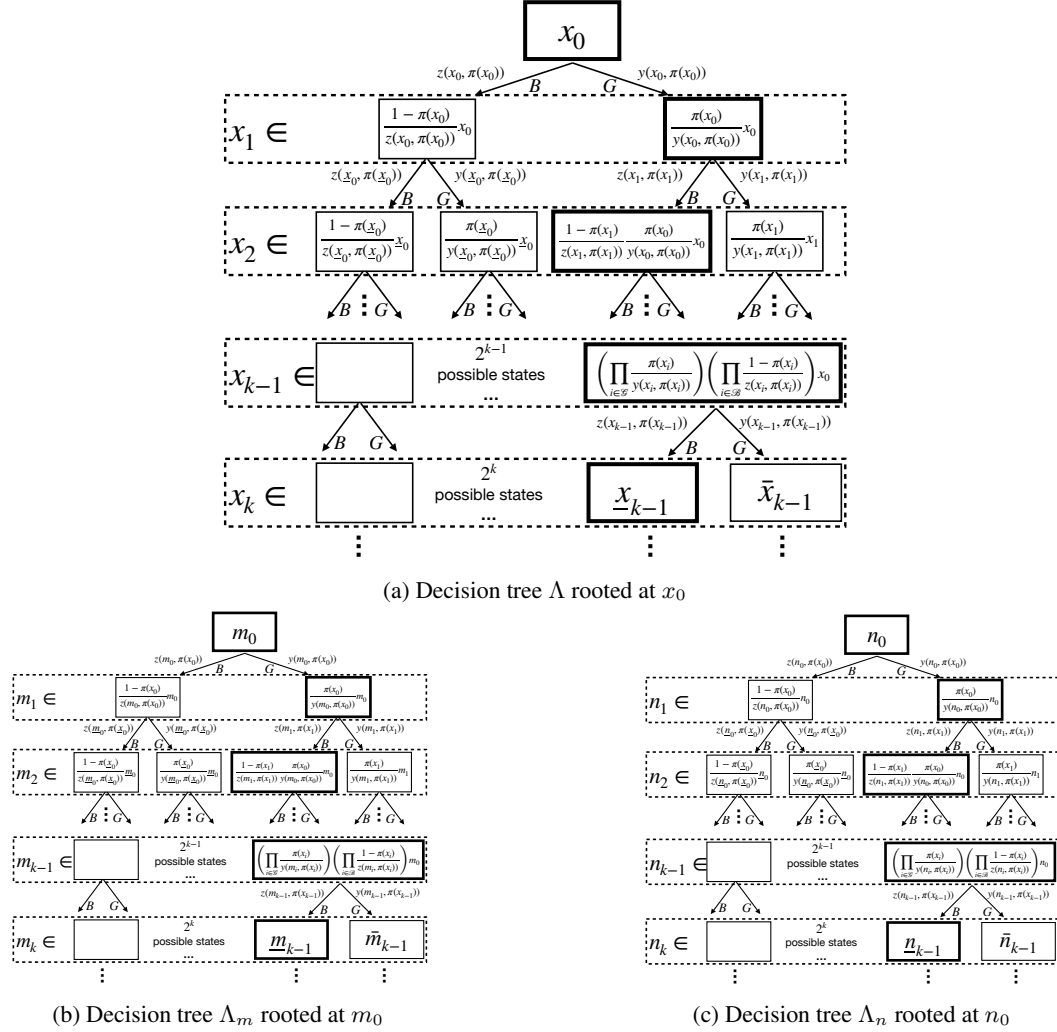


Figure 3: Here we depict the belief evolution in the form of three binary trees. The tree begins with state x_0 , m_0 , and n_0 , respectively, and level i contains 2^i possible states reachable after i epochs. Left branches correspond to action B leading to updated belief $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$). Right branches correspond to action G leading to updated belief $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with the corresponding probability, and the bold states indicate an example path, like λ . In the example path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states in each level of the three trees are counterparts.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^*}(x)$. Applying Fubini's Theorem, we can rewrite our value function

as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t \phi_i^{\tilde{\pi}}(m_0) + (1-t) \phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^\pi(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^*(x_0)$

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (29)$$

Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We also define $P_\lambda^{\tilde{\pi}}(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \leq t P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & + (1-t) P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \end{aligned} \quad (32)$$

Similarly, each pair of children for each term in the $(k-1)$ -th level of the belief tree can be combined and the corresponding inequalities can be obtained following the same process. Summing the resulting inequalities will yield (27), leading to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned}
1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
&+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)
\end{aligned} \tag{34}$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \tag{35}$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \tag{36}$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \tag{37}$$

We apply this as follows to the latter term of (29):

$$\begin{aligned}
&P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\
&\leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\
&\quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\
&= t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37})
\end{aligned} \tag{38}$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned}
&P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\
&\leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))]
\end{aligned} \tag{39}$$

Summing (38) and (39) yields (32).

We now complete the proof of (32) and, therefore, the entire proof, by proving (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we can write $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows. Each term in the products corresponds to the multiplicative state update after a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \tag{40}$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \tag{41}$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \tag{42}$$

These equations may be best understood via the illustration in figure 3. We then reorganize terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, using the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This proves (33).

Proof of (34)

We first consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this can be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 which depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ and the nature of the dependence is determined by the sequence of actions along this path in stages 1 to $k-1$. We will prove this via induction working backwards from the final term (i.e. the $(k-1)$ -th term) of the product. Therefore our base case will be the final term of this product which can be either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending on the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

Thus, the final term of the product is of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$ where α_{k-1} and β_{k-1} depend only upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is determined entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, assume that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ which depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the nature of the dependence is determined entirely by the sequence of actions in this path for $i > j$ to $i = k-1$. We will now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_j m_j + \beta_j$ again with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ with the nature of the dependence determined by the sequence of actions on

this path for $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$.

Thus,

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions in this path from $i > j$ to $i = k-1$. Thus, the claim holds for α_j, β_j , completing the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Note that the policy and sequence of actions is the same as those for the numerator. Thus, because, as shown, the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the path of actions, the same logic can be applied to write the denominator

as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . This can be done similarly for the numerator of the second term in (34). Thus, we can reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = t m_0 + (1-t) n_0) \\
& = 1
\end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Assume by way of contradiction that there exists $b \in [0, 1]$ such that $\pi_A^0(b) > \pi_A^*(b)$. We will show that $V_A^*(b) < V_A^\pi(b)$ for a policy π we construct, violating the optimality of $V_A^*(\cdot)$.

Let π be such that, when starting at public belief b , the policy π applies precision $\pi_A^0(b)$ at the current time step and then applies the optimal policy at all future time steps. We now elaborate $V_A^*(b) - V_A^\pi(b)$ by breaking the value function into instantaneous reward r_A and future discounted reward. We will refer to b^+ and b^- as the updated beliefs after receiving signals G and B , respectively when applying the optimal policy and b_m^+ and b_m^- as the same when applying the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

Here, we applied the fact that the instantaneous reward under $\pi_A^0(\cdot)$ must be greater from the definition of the myopic optimal policy.

We then relied on the convexity of $V_A^*(\cdot)$ from Theorem 2 for the future cost. From Equation (13), note that $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$. Along with this convexity, this leads to the last step above and completes the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ with policy π applying precision q in the current time step and applying the optimal policy at all future time steps. The q that maximizes $V_A^\pi(b)$ provides the optimum precision starting at public belief b . Because the planner never benefits from decreasing precision below the baseline p (see Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

We will let b^+ and b^- denote the positive and negative belief updates possible after receiving a signal with precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

With the convexity and concavity of $V_A^*(\cdot)$ and $\beta(\cdot)$, respectively, the above shows that $V_A^\pi(b)$ is convex with respect to q on $[\max(b, 1-b), 1]$. Thus, on this interval, it is maximized at one of the two extreme points.

For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative; thus, the optimal choice will be the baseline precision p to minimize the cost. Thus, there are three possible optimal precisions: p , $\max(b, 1 - b)$, and 1.

Note that the expected value under the latter two candidates is always strictly negative. Under precision p , however, the expected value is 0 at $b = 0$ and $b = 1$. This, combined with the non-positivity and symmetry of the value function (Lemma 8) implies the existence of d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A] \cup (1 - d_A, 1]$. Furthermore, from Theorem 1 and applying Lemma 9, $d_A \in (0, t_M)$ where t_M is defined as in the statement of Theorem 1.

Finally, Theorem 1 and applying Lemma 9, also imply the existence of $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule from (2), the instantaneous reward (6) can be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, $z(b, q) = z(0.5, q) = 0.5$ regardless of the precision chosen. Furthermore, at this public belief, $q \geq \max(b, 1 - b)$ because $q \geq 0.5$ for any q , falling in the first case of (44). In this case, the second term of $r_B(b, q)$ is unchanging with respect to q , so the planner seeks to maximize only the first term, which is accomplished at $q = p$.

We now prove a series of lemmas (10-14) which will assist us in our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except $q = p$. Thus, we can write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we can write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Because $\beta(\cdot)$ is concave, this expression is non-negative, and $r_B(b, q)$ is convex with respect to q . Thus, the reward is maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or p because $r_B(b, q)$ is not differentiable at $q = p$. □

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$, $\max(b, 1 - b) = 1 - b > p$.

First, consider the case when $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances, the reward is maximized by minimizing $|q - p|$. Thus, the optimal precision lower than $\max(b, 1 - b)$ is p .

Now consider $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$. What remains is to rule out $\pi_B^0(b) = 1$. To do so, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$. Both fall under the first case of (44). Because $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward from (44) is greater for precision $1 - b$ than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$ so we can rule out precision 1. This completes the proof. □

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The reward is strictly negative for any $q \neq p$ because of the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$, $p < \max(b, 1 - b) = b$, leading to the third case of (44) and yielding reward 0. Thus, the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ contains only $\{0.5\}$. At $b = 0.5$ $p = 1 - b$, thus $\pi_B^0(b) = p = 1 - b = 0.5$. We deal with $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$. Thus, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now consider two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This leads to the second case of (44) and yields reward $-\beta(|q - p|) - C$. Furthermore, $z(b, q)$ is increasing with respect to q for $b < 0.5$ and takes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$. Thus, $z(b, q) < 1$ for all q . Therefore, the reward from policy $q < 1 - b$ is strictly lower than $r_B(b, p)$, and we can rule out this case.

When $q \geq \max(b, 1 - b)$, applying Lemma 10 yields that the remaining candidate precisions are $\{1 - b, p, 1\}$. We rule out precision 1 by comparing it to precision p . The first term of the reward from (44) is greater for precision p than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, the second term is also greater for precision p , and we can rule out precision 1. This completes the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that from the definition, any optimal policy is also ϵ -optimal for any $\epsilon > 0$. Thus, $\pi_B^0(b)$ is also ϵ -optimal

Following this definition, let $\pi_B^{\epsilon,0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or

2. $\pi_B^{\epsilon,0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. We consider two cases: $q \geq \max(b, 1 - b)$ or $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields that $\pi_B^0(b) \in \{b, p, 1\}$. We rule out precision b by comparing it to precision p . The first terms of the reward from (44) is greater for precision p . Furthermore, since $z(b, q)$ is decreasing with respect to q for $b > 0.5$, the second term of the reward is also greater for precision p . Thus, we can rule out precision b . This yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Therefore, (1) since $q \geq 0.5$, $q \in [0.5, b)$, and (2) the third case of (44) applies leading to reward $-\beta(|q - p|)$. Thus, the reward $r_B(b, q)$ is an increasing function of $q \in [0.5, b)$ because $0.5 < b \leq p$. This leads directly to the second claim. $\square$

Remark 3. As noted in Puterman (1990) Section 2.3.1, an optimal policy may not exist for infinite action spaces because the supremum may not be attained. When an optimal policy does not exist, we instead seek ϵ -optimal policies, i.e., policies that yield reward within $\epsilon > 0$ of the supremum. In particular, this is true for case 2 of Lemma 14.

Proof of Theorem 4 We consider the myopic biased policy in 4 cases: $b \in [0, 1 - p]$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$, making use of the fact that $p \geq 0.5$. Based on Lemmas 11-14, these cases have the following candidate policies: $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We will show the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

We can express the reward under the two candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive as $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Furthermore, when evaluated at $b = 0$ and $b = 1 - p$ we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus, $r_B(b, 1 - b)$ is increasing on $[0, 1 - p]$ with $r_B(b, 1 - b) < r_B(b, p)$ for $b = 0$ and vice versa for $b = 1 - p$. Because $\beta(\cdot)$ is continuous, so are both reward functions. Thus, there exists $t_1 \in (0, 1 - p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, this interval contains only $\{0.5\}$, and, for $b = 0.5$, $1 - b = p$. Thus, the two candidate policies are equivalent, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now prove the existence of t_2, t_3 such that $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3]$, and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1 - b$:

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1 - p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1 - p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1 - p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1 - p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1 - b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, and, therefore, $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b - \epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b - \epsilon$ is the optimal policy, must also be convex. Therefore, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b can be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, note that the Bayesian nature of belief updates implies that the belief process is a martingale. That is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, it follows from (3) that $b_{i+1} = b_i$, satisfying the claim.

In the latter case, (13) gives the distribution of b_{i+1} which yields the same.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. Biased value function continuity

$V_B^*(\cdot)$ is continuous w.r.t. b .

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1 - b, 1, b - \epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1 - b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of

(44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_\epsilon < p$ such that on $[t_\epsilon, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ + \delta y(b, q) V_B^*(f(b, q, G)) \\ + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ -C\mathbf{1}(b < 0.5) \\ + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p]$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1-b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q-p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1-b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1-\epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q-p|) \rightarrow \beta(1-p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1-p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p-p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1-p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1-p, 0.5)$, $\pi^*(b_i) \geq 1-b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1-b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1-b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1-\delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1-\delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1-b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1-b_i$.

The remaining candidate policies are $q_i \in [1-b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1-b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1-b_i)$ and $b \in [1-p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1-q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1-b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1-b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the

supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \tag{57}$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \tag{58}$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is clear that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Furthermore, note that the convexity of social welfare with respect to public belief follows from the same argument as in the proof of Theorem 2. This is because social welfare is equivalent to the altruistic value function in the special case where $\beta(\cdot) = 0$.

We can then follow the same argument as in the proof of Lemma 9 (Appendix C.4) to show that increasing precision increases welfare. $\square$

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

We show that, for the altruistic planner, the same core problem extends to an environment with two observable agent types that have opposing preferences.

Assume there are two agent types, G and B , and let t_i denote agent i 's type. The first type receives unit reward when $a_i = \omega$ and no reward otherwise; the second receives unit reward when $a_i \neq \omega$ and no reward otherwise. All agents know their own type and the types of all other agents, while the rest of the model is unchanged.

Agent i 's private belief update is unchanged because it depends only on the signal structure, i.e., $\mathbb{P}(s_i = \omega) = q_i$:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_iq_i-b_i-q_i} b_i & s_i = G \\ \frac{1-q}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), agent i 's action can be written as follows, where the superscript indicates type and $!$ denotes negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because each agent's type is public, all agents can perfectly infer agent i 's private signal whenever $q_i \geq \max(b_i, 1 - b_i)$. If $q_i < \max(b_i, 1 - b_i)$, agent i 's action is uninformative.

Therefore, after observing type t_i , precision q_i , and action a_i , public belief updates as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this heterogeneous setting, an altruistic planner induces type- G agents to take action ω and type- B agents to take action $!\omega$, because these choices maximize each type's own utility.

The planner's instantaneous reward can then be expressed as follows, with the superscript denoting agent i 's type:

$$\begin{aligned} r^G(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega) \\ &= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \\ r^B(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega) \\ &= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \end{aligned}$$

The two expressions are identical and equal to the reward function in Equation (5), establishing equivalence with the altruistic-planner problem for heterogeneous but observable types. $\blacksquare$

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To simulate controlled social learning, we use a sequence of agents deciding whether to buy a new car model after observing past actions and receiving an ad whose precision is selected by the planner. LLMs serve three

roles: (1) agents, (2) planner, and (3) oracle. Each agent receives a profile containing demographic information, car-use cases, and preferences. At the outset, the offered car is fixed as either good or bad. The information sets and objectives of the planner and agents match the model in Section 3.

The oracle has two roles: it generates a generic message that serves as the baseline-precision signal without planner intervention, and, after the planner chooses a precision, it generates a profile-customized message with that prescribed precision.

The oracle receives a fact sheet for each car type, including attributes such as safety rating and customer satisfaction, together with the agent profile. The fact sheet and profile jointly support generation of a tailored message at the planner’s requested precision. Appendix E.2 provides additional setup and prompting details.

Each agent makes a binary decision using previous decisions and her private signal, with the goal of matching the true state.

The planner observes the same decision history and chooses a precision level for each agent’s message. It also knows the precision of a generic, untailored message that can be sent at zero cost. Under the altruistic objective, the planner seeks correct decisions; under the biased objective, it seeks to induce agents to buy the car.

We compare the LLM simulations with analytical solutions obtained via JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

Experiments use the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the model analytically and run the LLM simulations for the parameter grid in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	{0.1, 0.3, 0.5, 0.7, 0.9}
p	Baseline signal precision	{0.6, 0.7, 0.8, 0.9}
δ	Discount Factor	{0, 0.25, 0.5, 0.75, 1}

Table 1: Parameter grid used for analytical and LLM experiments.

The car fact sheets are constructed so that the GOOD car is beneficial for every profile and the BAD car is harmful for every profile. The fact sheets for GOOD and BAD cars are listed below.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100→72 °F in 3 min.",
    "Cargo volume 26 ft³ total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
    "EPA-rated range: 155 mi (combined).",
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
    "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
    "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%.",
    "Cabin cools 105 °F → 72 °F in 11 min.",
    "Cargo volume 17 ft³; rear seats fixed (no fold-flat).",
    "No factory towing approval.",
    "Average 3.8 service visits per year (battery-alert issues).",
    'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
    "Not eligible for federal tax credit.",
]
```

```

1836         "OTA updates limited to infotainment."
1837     ]
1838

```

Agent profiles are generated programmatically and include name, age, city of residence, interests, planned car use cases (e.g., commuting or road trips), and major concerns (e.g., fuel efficiency or winter driving).

E.2.2 PROMPTS

Agent LLM Prompt:

```

1845 sys = (
1846     f"You are deciding whether to buy a car. "
1847     f"Your profile is {profile}"
1848     f"You start believing there's a 50% chance the car is GOOD. "
1849     f"You have two information sources: "
1850     f"1. Previous agents' decisions "
1851     f"2. An advisor's message about the car "
1852     f"Your goal is to maximize your reward "
1853     f"i.e. BUY if the car is GOOD and "
1854     f"NOT BUY if the car is BAD. "
1855     f"If you BUY a GOOD car, you GAIN ${C}. "
1856     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1857     f"If you BUY a BAD car, you LOSE ${C}. "
1858     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1859     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1860     f"decision = exactly 'BUY' or 'NOT BUY'."
1861 )
1862 user = (
1863     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1864     f"ADVISOR MESSAGE:\n{message}\n\n"
1865     f"TASK: Use the previous decisions and the advisor message. "
1866     f"Decide BUY or NOT BUY. "
1867     f"Update your belief about car quality (0.0-1.0). "\n"
1868     f"JSON response:{{'belief': <number>,'decision': '<BUY or NOT BUY>'}}"
1869 )

```

Planner LLM Prompt:

```

1868 sys = (
1869     f"You are providing messages to potential car buyers. "
1870     f"Each potential car buyer (or agent) has the same information. "
1871     f"Your goal is to get agents to {objective}."
1872     f"Each decision according to your objective, earns you {C} dollars. "
1873     f"You and the agents have the same information. "
1874     f"Initially there is a 50% chance the car is GOOD. "
1875     f"The agents' aim to BUY if the car is GOOD "
1876     f"and NOT BUY if the car is BAD. "
1877     f"You can send a baseline message for free "
1878     f"or tailor the message to the agent at a cost. "
1879     f"Tailoring costs {k} dollars times the change in precision. "
1880     f"(Message precision is the probability (0.5 to 1.0) "
1881     f"that the message leaves the buyer with the correct impression. "
1882     f"For example, if the car is likely GOOD: "
1883     f"a high precision message will likely suggest BUY "
1884     f"and a low precision message will likely suggest NOT BUY. "
1885     f"You may choose to increase or decrease precision. "
1886     f"Agents decide based on your message and the decision history. "
1887     f"They update beliefs by observing previous decisions. "
1888     f"IMPORTANT: Thus your objective is to maximize:"
1889     f"{C} * (# of agents who choose according to your objective) "
1890     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1891     f"where {d} is your discount factor for future cost. "
1892 )
1893 user = (
1894     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"

```

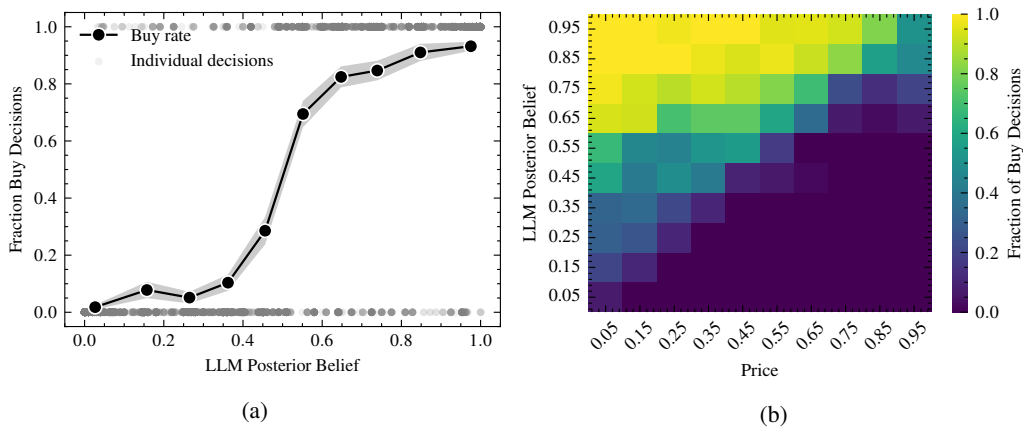
```

1890     f"DECISION HISTORY:\n{history_txt}\n"
1891     f"TASK:"
1892     f"Based on the history and your objective, "
1893     f"choose your desired precision.\n"
1894     f'JSON response: {{"message_precision": <number>}}\n'
1895 )
1896
1897 Oracle LLM Prompt:
1898
1899 sys = (
1900     f"You will be given "
1901     f"1. an agent profile, "
1902     f"2. information about a car, "
1903     f"3. the true quality of the car, "
1904     f"4. a baseline message and the 'precision' of that message, and "
1905     f"5. a desired precision for the message you output. "
1906     f"(Message precision is the probability (0.5 to 1.0) "
1907     f"that the message leaves the buyer with the correct impression. "
1908     f"For example, if the car is likely GOOD: "
1909     f"a high precision message will likely suggest BUY "
1910     f"and a low precision message will likely suggest NOT BUY. "
1911     f"IMPORTANT: Output a message with the requested precision. "
1912     f"IMPORTANT: The message should be tailored to the profile. "
1913 )
1914
1915 user = (
1916     f"AGENT PROFILE:{profile}\n"
1917     f"TRUE CAR QUALITY: {omega}\n\n"
1918     f"CAR FACTS: {car_facts}\n"
1919     f"BASELINE MESSAGE:{baseline_message}}\n"
1920     f"BASELINE MESSAGE precision: {p}\n"
1921     f"DESIRED MESSAGE precision: {q}\n"
1922     f"IMPORTANT: Output a message with the requested precision. "
1923     f"TASK: Write a message (MAX 100 words) with the desired precision."
1924     f'JSON response: {{"message": "<message>}}\n'
1925 )

```

1921 E.3 VALIDATION OF LLM BELIEFS AND ORACLE

1922 We provide additional experiments validating both the self-reported LLM beliefs used in figure 1b and the LLM oracle's message precision. These checks treat LLMs as rational, utility-maximizing agents that need not update as Bayesians. The goals are (1) to show that the model's self-reported posterior corresponds to the belief governing its choices in state-contingent decision problems and (2) to verify that oracle-generated messages achieve the requested precision.



1942 Figure 4: LLM buy-decision rates as a function of (a) self-reported posterior belief and (b) self-reported posterior together with the price of buying.

First, we give the LLM a prior and a new signal, elicit its self-reported posterior belief, and then ask for a binary “buy/not-buy” decision that yields unit reward if and only if the true state is good. A rational agent should choose “buy” exactly when its subjective belief exceeds 0.5. As shown in figure 4a, the LLM’s decisions align sharply with this threshold: it overwhelmingly chooses “buy” when its reported posterior exceeds 0.5 and “not buy” otherwise. Next, we elicit the model’s willingness to pay for a payoff contingent on the true state. For a range of prices c , we ask whether the agent would pay c in exchange for unit utility if the state is good. A rational agent with belief p should accept exactly when $c < p$. Figure 4b plots the fraction of “buy” decisions across posterior–price pairs and shows a clear separation: acceptance rates are high when $c < p$ and low when $c > p$, as expected from a utility-maximizing decision maker whose belief is p . Together, the experiments indicate that the self-reported posterior closely tracks the belief used by the model in decision-making.

We also validate the oracle’s ability to craft messages at a specified precision. After the oracle generates a message with requested precision q , we ask another LLM to estimate the message precision $\hat{q}$. We evaluate $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ on 100 programmatically generated agent profiles, with 10 trials for each combination, for a total of 6,000 trials. The mean absolute error $|q - \hat{q}|$ is 0.025, indicating that realized oracle-message precision closely follows requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We finally examine belief trajectories from an example LLM experiment to illustrate how non-Bayesian updating affects aggregate learning. We fix the policy to the analytically optimal policy and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0. The LLM belief (solid black line) is more persistent. This macro-level trajectory follows from the micro-level biases: public belief resists entering a cascade (NB3) because individual agents overreact to occasional positive signals that contradict the group’s increasingly negative belief (NB2).

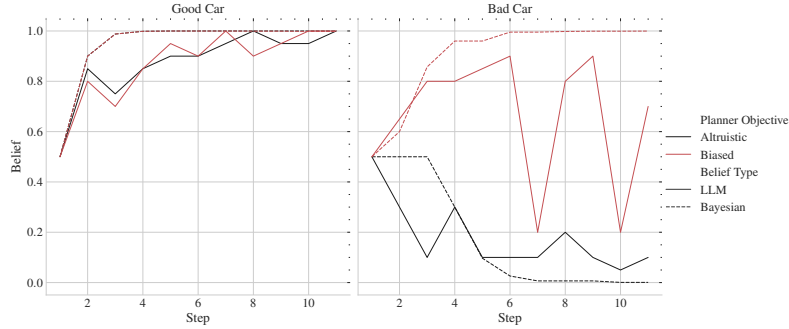


Figure 5: Belief-updating trajectories for the Bayesian public belief and the belief reported by LLM agents under analytically optimal policies, across planner objectives and true states.